

End-to-end LST workflow in pyMack

1 · Flow conditions

M, Re, T_w, gas

2 · Base (mean) flow

compressible Blasius BVP $\rightarrow \bar{U}, \bar{T}$

3 · Linearise + normal mode

ODEs in y for $\hat{u}, \hat{v}, \hat{T}, \hat{p}$

4 · Chebyshev collocation

D_1, D_2 on the wall-clustered grid

5 · Eigenvalue solve

the heart of LST

TEMPORAL

$$A\phi = cB\phi$$

growth = αc_i

Sweep (R, α)

build c_i field

Neutral curve

$\{c_i = 0\}$ + continuation

SPATIAL

$$(C_0 + \alpha C_1 + \alpha^2 C_2)\phi = 0$$

growth = $-\alpha_i$

$$N = \int -\alpha_i dR$$

$e^N \rightarrow$ transition